

Question number	Answer	Mark
5(a)	sequence of bases of DNA that code for a polypeptide	(1)

Question number	Answer	Mark
5(b)	A description that includes the following points: - complementary bases / named pair of complementary bases (1) - it enables the formation of {two hydrogen bonds between adenine and thymine / three hydrogen bonds between cytosine and guanine} (1)	(2)

Question number	Answer	Additional guidance	Mark
5(c)	A explanation that includes the following points: - enough codons needed for 20 different amino acids (1) - four bases are used in the genetic code (1) (triplet code) provides {enough / 43 / 64} possible codons (1)	Allow descriptions of single and doublet code providing insufficient alternatives	(3)

Question number	Answer	Additional guidance	Mark
5(d)	A answer that includes the following points: - correct genotypes of parents (1) - affected genotype of children correctly identified (1) - correct calculation of probability is 0.5 (1)	Accept 50%, 1 in 2, $\frac{1}{2}$	(3)